BUSINESS CUSTOMER RISK CLASSIFICATION

A Machine Learning Approach to High-Risk Customer Identification

Technical Report

Prepared using: Logistic Regression · Decision Tree · Random Forest

Dataset: 5,000 Customer Records

August 2026

Executive Summary

This report presents a complete machine learning workflow for classifying business customers into Low Risk and High Risk categories. Using a dataset of 5,000 customer records, three algorithms — Logistic Regression, Decision Tree, and Random Forest — were developed and compared across Accuracy, Precision, Recall, F1-score, and ROC-AUC.

A target leakage investigation identified customer_index as a variable that had been used to construct the target itself, and it was therefore excluded from the final feature set. The remaining predictors — customer age, monthly spend, number of visits, and number of complaints — were used to train the final models.

Logistic Regression delivered the strongest and most reliable performance, achieving 99.8% test accuracy, 99.8% precision, 99.8% recall, and an ROC-AUC of approximately 100%, with only two misclassified customers out of 1,000 in the test set. Random Forest was a strong second-place model, while Decision Tree showed the greatest evidence of overfitting.

Because the dataset is synthetic and the target variable was mathematically constructed, the exceptionally high accuracy should be interpreted with caution. The recommended next step is to validate the selected model against real, independent customer data before any production deployment.

1. Introduction

Businesses often need to identify customers who may require additional attention because of their risk characteristics. Early identification of high-risk customers helps an organization improve customer management, prioritize interventions, and allocate resources more effectively.

This project focuses on developing a machine learning classification system for identifying customers who belong to a High Risk category. The task was implemented using a complete data preparation and machine learning workflow, including data inspection, exploratory analysis, leakage investigation, feature selection, model development, model evaluation, and interpretation.

Three machine learning algorithms were investigated: Logistic Regression, Decision Tree, and Random Forest. Their performance was compared using Accuracy, Precision, Recall, F1-score, ROC-AUC, and confusion matrices.

2. Business Problem and Machine Learning Formulation

2.1 Business Problem

The business wants to identify customers who belong to a high-risk category before taking appropriate business action. The objective is to develop a predictive model that classifies each customer into one of two categories:

- 0 — Low Risk
- 1 — High Risk

The model learns patterns from customer characteristics such as age, spending behavior, number of visits, and complaints.

2.2 Machine Learning Problem Definition

This problem was formulated as a supervised binary classification problem. Supervised learning means the algorithm learns from historical data where the correct outcome (the target variable) is already known. In this project, the pipeline follows the structure: input features flow into a machine learning model, which produces a High Risk or Low Risk prediction.

The target variable was high_risk, where 0 represents Low Risk and 1 represents High Risk. The predictor variables initially available were customer_age, monthly_spend, visits, complaints, and customer_index. customer_index was later excluded from the final modeling dataset because of a potential target leakage issue (see Section 6).

3. Dataset Description

The dataset contained 5,000 customer records and 6 columns, consisting of five numerical input/constructed variables and one binary target variable.

Variable	Description
customer_age	Age of the customer
monthly_spend	Customer's monthly spending
visits	Number of customer visits
complaints	Number of complaints
customer_index	Constructed customer score
high_risk	Target classification

Table 1. Dataset variables and descriptions

The final target distribution was balanced, with 50% Low Risk and 50% High Risk customers in both the training and testing datasets.

4. Data Inspection and Preparation

4.1 Missing Values

The dataset was examined for missing values. All six variables contained zero missing observations (total missing values = 0). This simplified the preprocessing stage, as no missing-value imputation using methods such as mean or median substitution was required.

4.2 Duplicate Records

Duplicate records were also investigated, and none were found (number of duplicate rows = 0). Therefore, no duplicate records were removed from the dataset.

5. Exploratory Data Analysis

5.1 Descriptive Statistics

Customer age ranged from 18 to 79 years, with a mean of approximately 48.45 years. The number of visits ranged from 1 to 79, with a mean of approximately 39.90 visits. Complaints ranged from 0 to 10, with an average of approximately 2.02 complaints.

The most unusual variable was monthly_spend, which showed a strongly right-skewed distribution:

Statistic	Monthly Spend
Mean	7,946.089
Median	396.835
Minimum	0.069
Maximum	1,102,074.941
Standard deviation	44,833.530

Table 2. Descriptive statistics for monthly_spend

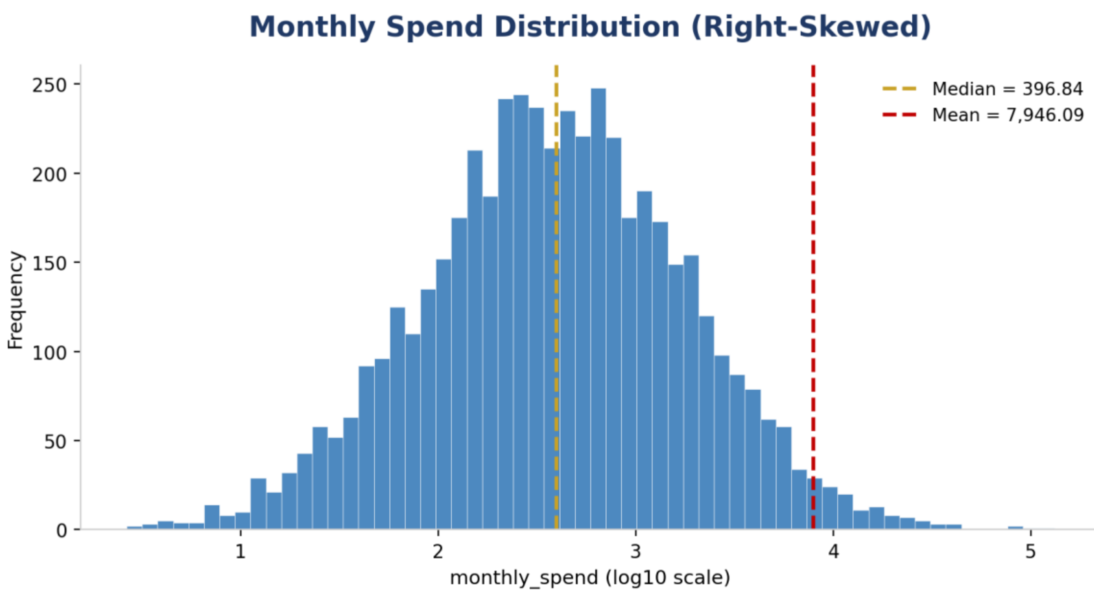

Figure 1. Monthly spend distribution — the large gap between the mean and median confirms the right-skew.

5.2 Target Variable Distribution

The target variable was evenly distributed across both the training and testing datasets (50% Low Risk / 50% High Risk). This balance is important because neither class was severely underrepresented, meaning the model did not need special class-balancing techniques such as SMOTE solely because of target imbalance.

6. Feature Relationships and Target Leakage Investigation

6.1 Correlation Analysis

Correlation analysis was performed to understand the relationship between the available numerical variables and the target. The strongest relationship was observed for customer_index, followed closely by visits. Correlation alone does not establish causation — it only indicates the strength and direction of a statistical relationship.

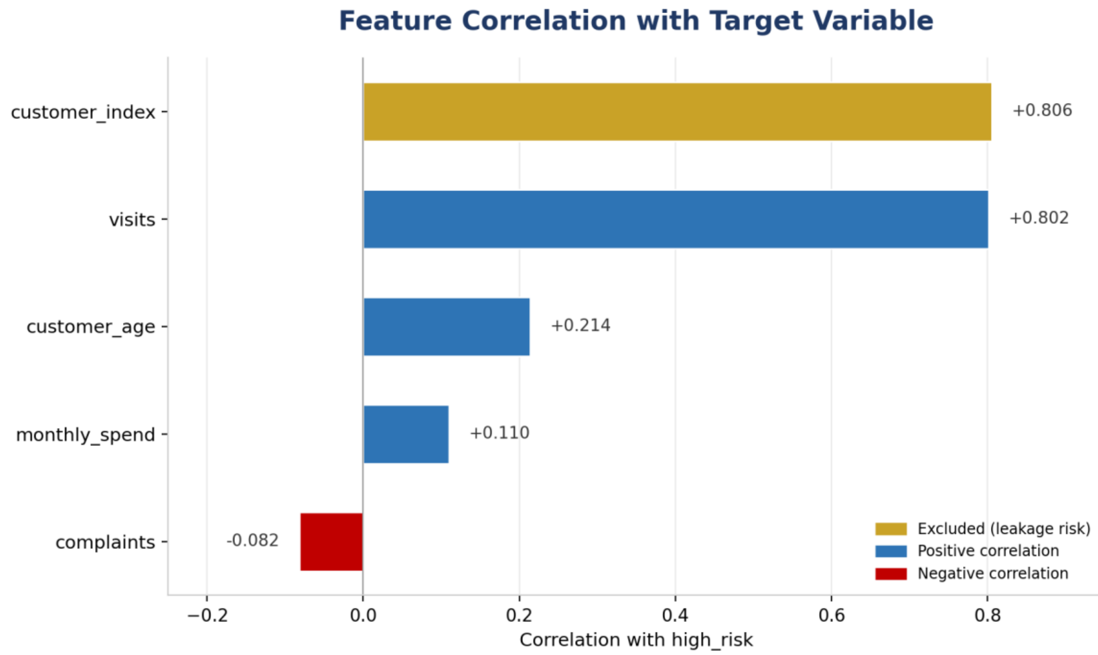

Figure 2. Correlation of each feature with the high_risk target variable.

6.2 What Is Target Leakage?

Target leakage occurs when an input feature contains information directly or indirectly derived from the target — information the model would not legitimately have when making a real-world prediction. In this dataset, high_risk was created using a threshold applied to customer_index: the median customer_index was approximately 2.48094, and customers above this threshold were assigned to the High Risk category. Analysis confirmed that exactly 2,500 customers fell at or below the median and 2,500 fell above it.

6.3 Decision on customer_index

Although customer_index had a strong correlation of approximately 0.806 with high_risk, this relationship was partly expected because the target was constructed from that variable. Including it would make the classification problem unrealistically easy. Therefore, customer_index was excluded from the final feature set, leaving customer_age, monthly_spend, visits, and complaints as the final predictors.

7. Train-Test Split

The dataset was divided into training and testing subsets using an 80:20 split, resulting in 4,000 training observations and 1,000 testing observations, each with four features after customer_index was excluded.

The target distribution was preserved using stratification, resulting in an equal 50/50 class distribution in both datasets. The training data were used to fit the models, while the test data were kept separate for evaluating generalization to unseen observations.

8. Machine Learning Algorithms and Results

Three algorithms were selected for comparison: Logistic Regression, Decision Tree, and Random Forest. Each represents a different modeling approach, ranging from a linear probabilistic model to ensemble tree-based methods.

8.1 Logistic Regression

Logistic Regression is a classification algorithm commonly used for binary outcomes. Instead of directly predicting a numerical value, it estimates the probability that an observation belongs to a particular class — in this case, the probability that a customer belongs to the High Risk category.

Metric	Performance
Training Accuracy	99.92%
Test Accuracy	99.80%
Precision	99.80%
Recall	99.80%
F1-score	99.80%
ROC-AUC	100.00%

Table 3. Logistic Regression performance

The model correctly classified 499 Low Risk and 499 High Risk customers, with only two observations incorrectly classified. The training-test accuracy gap was only 0.12 percentage points.

8.2 Decision Tree

A Decision Tree classifies observations through a sequence of decision rules and is capable of capturing non-linear relationships and interactions between variables.

Metric	Performance
Training Accuracy	100.00%
Test Accuracy	96.40%
Precision	96.59%
Recall	96.20%
F1-score	96.39%
ROC-AUC	96.40%

Table 4. Decision Tree performance

The training accuracy reached 100%, but test accuracy fell to 96.4% — an accuracy gap of 3.6 percentage points, providing clear evidence of overfitting.

8.3 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple Decision Trees to produce a more robust prediction, rather than relying on a single tree.

Metric	Performance
Training Accuracy	100.00%
Test Accuracy	97.80%
Precision	97.42%
Recall	98.20%
F1-score	97.81%
ROC-AUC	99.87%

Table 5. Random Forest performance

Random Forest showed a 2.2 percentage-point training-test accuracy gap — smaller than the Decision Tree's gap but larger than Logistic Regression's.

9. Model Performance Comparison

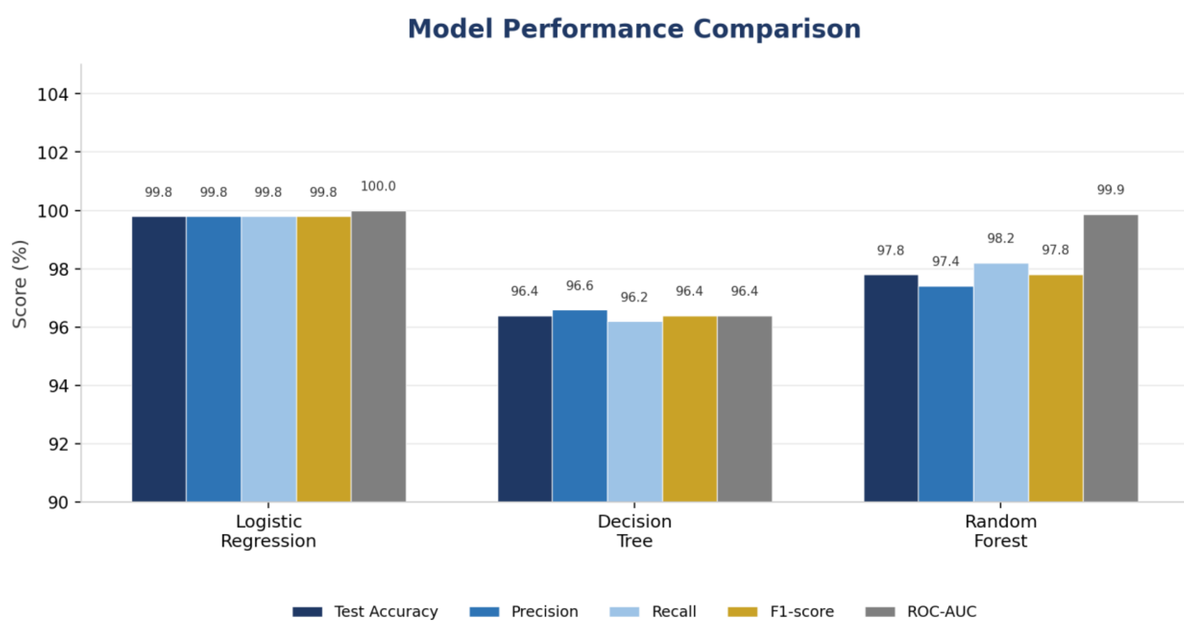

Figure 3. Side-by-side comparison of all evaluation metrics across the three models.

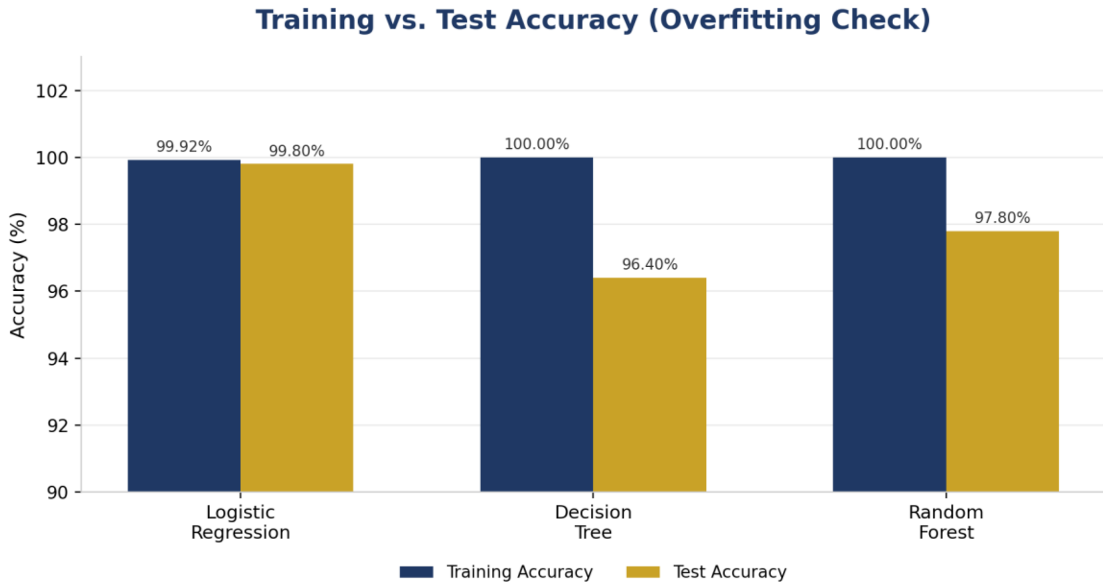

Figure 4. Training vs. test accuracy — the gap indicates the degree of overfitting for each model.

Logistic Regression achieved the highest test accuracy, precision, recall, F1-score, and ROC-AUC. Random Forest was the second-best model, achieving a particularly strong recall of 98.2%. Decision Tree produced the lowest test performance and the largest train-test gap.

Overall ranking based on the recorded results:

- 1st — Logistic Regression
- 2nd — Random Forest
- 3rd — Decision Tree

10. Confusion Matrix and Error Analysis

Confusion matrices were used to investigate the types of errors made by each model, distinguishing True Positives (High Risk correctly identified), True Negatives (Low Risk correctly identified), False Positives (Low Risk incorrectly flagged as High Risk), and False Negatives (High Risk incorrectly classified as Low Risk).

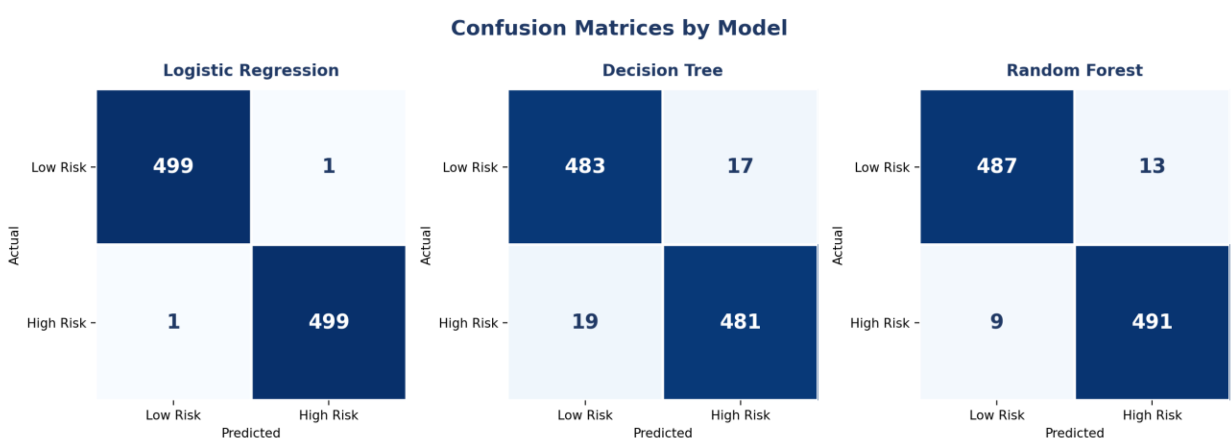

Figure 5. Confusion matrices for Logistic Regression, Decision Tree, and Random Forest.

Logistic Regression produced the fewest errors overall (1 false positive, 1 false negative). Random Forest produced fewer false negatives than the Decision Tree (9 vs. 19), which is important from a business perspective since a false negative represents a genuinely high-risk customer who was missed by the model.

10.1 Evaluation Metrics Explained

- Accuracy — the proportion of all predictions that were correct.
- Precision — how many customers predicted as High Risk were actually High Risk.
- Recall — how many actual High Risk customers were successfully identified.
- F1-score — the harmonic balance between precision and recall.
- ROC-AUC — how effectively a model separates the two classes across different thresholds.

For this business problem, recall is particularly important because failing to identify a genuinely high-risk customer may carry direct business consequences.

11. Feature Importance and Interpretation

Feature importance was investigated for the tree-based models. In both cases, visits was clearly the most influential predictor, consistent with the correlation analysis, where visits showed a strong positive relationship with high_risk.

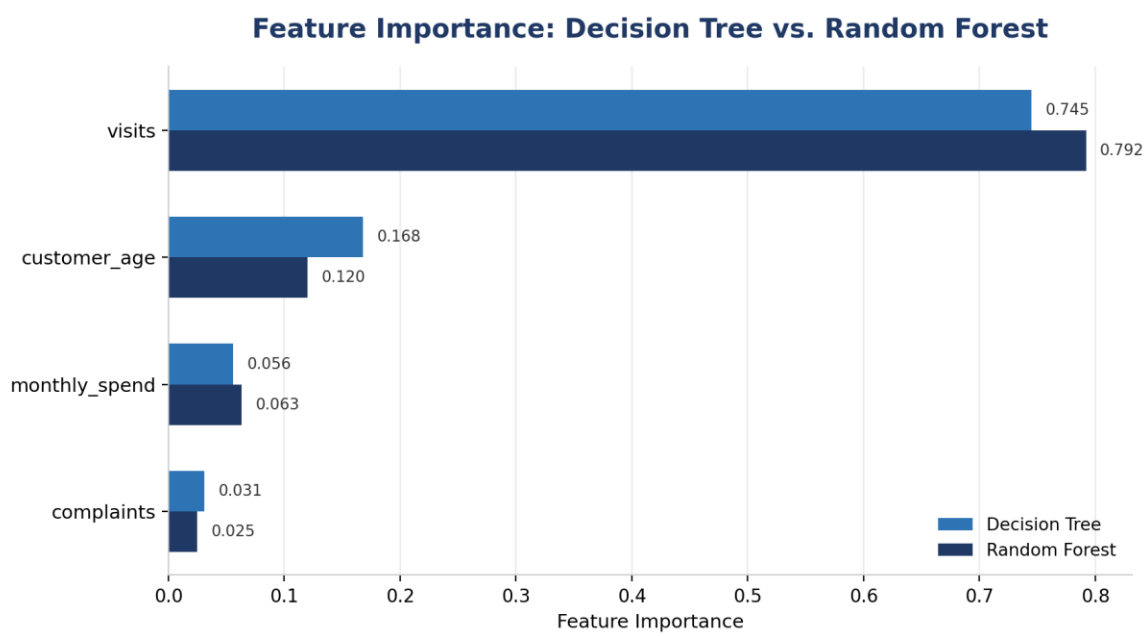

Figure 6. Feature importance comparison between Decision Tree and Random Forest.

The Decision Tree relied on visits for approximately 74.5% of its total feature importance, while Random Forest relied on it for approximately 79.2%. In both models, customer_age was the second most influential feature, followed by monthly_spend and complaints.

12. Investigation of Unusually High Performance

The models produced unusually high performance, particularly Logistic Regression, which reached 99.8% test accuracy and an ROC-AUC of approximately 1.00. Such results should be interpreted carefully.

The primary explanation lies in the nature of the dataset itself: the target variable was synthetically generated from customer characteristics using a mathematical rule, making the classification boundary relatively structured and predictable. The results should therefore not be interpreted as evidence that an equivalent model would necessarily achieve 99.8% accuracy on real-world customer data.

13. Final Model Selection

Based on the recorded experimental results, Logistic Regression was selected as the final model, for the following reasons:

- Highest test accuracy: 99.8%
- Highest precision: 99.8%
- Highest recall: 99.8%
- Highest F1-score: 99.8%
- ROC-AUC of approximately 100%
- Very small training-test accuracy gap of 0.12 percentage points
- Only two test-set classification errors out of 1,000

Random Forest was a strong alternative, with a test accuracy of 97.8% and recall of 98.2%, but it did not outperform Logistic Regression in the recorded experiment. Decision Tree was not selected because it showed the largest generalization gap and the lowest test performance.

13.1 Model Saving and Deployment

The selected Logistic Regression model was saved using the joblib library (joblib.dump) as group5_logistic_regression_model.pkl, and can later be reloaded (joblib.load) for integration into an application such as a web-based prediction system that accepts Customer Age, Monthly Spend, Number of Visits, and Number of Complaints, and returns a Low Risk or High Risk classification.

14. Limitations

14.1 Synthetic Dataset

The dataset was generated synthetically rather than collected from real customers. Therefore, the extremely high predictive performance may not generalize to real-world customer behavior.

14.2 Target Construction

The target variable was generated using a mathematical rule based on the customer characteristics, making the classification problem more structured than many real-world business problems.

14.3 Potential Leakage

customer_index was strongly associated with the target because it was involved in the target construction process, and was therefore excluded from the final feature set.

14.4 Extreme Monthly Spending Values

monthly_spend contained very large extreme observations that should be investigated further using genuine business data to determine whether they represent valid high-value customers or data-quality problems.

14.5 Incomplete Recorded Experiments

The available experiment record does not provide quantitative results for the effect of normalization, outlier removal, cross-validation, or hyperparameter tuning. These should not be presented as completed quantitative experiments without running the corresponding analyses.

15. Conclusion and Recommendations

This project demonstrated a complete machine learning workflow for Business Customer Classification. The dataset consisted of 5,000 customer records with no missing values or duplicate observations. Exploratory analysis showed that monthly_spend was highly skewed with extreme values, while visits demonstrated a strong relationship with the target variable.

A target leakage investigation identified customer_index as a potentially problematic feature, since the target variable was constructed using this index, and it was excluded from the final predictive features. Among the three algorithms evaluated, Logistic Regression achieved the strongest overall performance (99.8% test accuracy, precision, recall, and F1-score, with ROC-AUC of approximately 100%). Random Forest also performed strongly, while Decision Tree showed the largest evidence of overfitting.

Overall, Logistic Regression was selected as the final model because it provided the best combination of predictive performance, generalization, and simplicity. However, the exceptionally high performance must be interpreted cautiously: since the dataset is synthetic and the target was mathematically constructed, the results may not represent performance on real-world customer data.

15.1 Recommended Workflow for Future Implementation

- Collect and validate real customer data
- Perform data quality checks
- Conduct a leakage investigation
- Carry out feature engineering
- Apply a train/test split with stratification
- Perform cross-validation
- Conduct hyperparameter tuning
- Compare candidate models
- Perform error analysis
- Select the final model
- Deploy the model into production

- Establish continuous monitoring

The most important recommendation is not to rely solely on the 99.8% accuracy obtained from this synthetic dataset. The model should first be tested on real, independent customer data to determine whether it can generalize reliably in a genuine business environment.